SUBJECT TEACHING GUIDE

G1054 - Loading and Stevedoring

Degree in Nautical Engineering and Maritime Transport

Academic year 2025-2026

1. IDENTIFYING DATA					
Degree	Degree in Nautical Engineering and Maritime Transport			Type and Year	Compulsory. Year 3
Faculty	School of Maritime Engineering				
Discipline	Subject Area: Cargo Handling and Stowage				
Course unit title and code	G1054 - Loading and Stevedoring				
Number of ECTS credits allocated	6	Term		Semester based (2)	
Web	https://ocw.unican.es/course/view.php?id=306				
Language of instruction	Spanish	English Friendly	No	Mode of delivery	Face-to-face

Department	DPTO. CIENCIAS Y TECNICAS DE LA NAVEGACION Y DE LA CONSTRUCCION NAVAL
Name of lecturer	FRANCISCO JOSE CORREA RUIZ
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Other lecturers	FRANCISCO VUELTA FERNANDEZ

3.1 LEARNING OUTCOMES

- Cargo handling, stowage & securing:
  Knowledge of the effect of cargo including heavy lifts on the seaworthiness and stability of the ship .
  Knowledge of safe handling, stowage and securing of cargoes including solid bulk cargoes and dangerous, hazardous and harmful cargoes and their effect on the safety of life and of the ship .
  Ability to establish and maintain effective communications during loading and unloading .
  Cargo handling equipment. The correct rigging and safe operation of derricks, cranes, and stores hoists.
  Crude oil tankers, products tankers, gas tankers and chemical tankers: Knowledge of cargo equipments, measure and control of tanks atmosphere.

4. OBJECTIVES

Plan and ensure safe loading, stowage, securing, care during the voyage and unloading of cargoes
Monitor the loading, stowage, securing, care during the voyage and the unloading of cargoes
Assess reported defects and damage to cargo spaces, hatch covers and ballast tanks and take appropriate action
Inspect and report defects and damage to cargo spaces, hatch covers and ballast tanks
Carriage of dangerous goods
Contribute to the safe cargo operation of oil, chemical and gas tankers. Take precautions to prevent hazards. Apply occupational health and safety precautions and measures. Ability to safely perform and monitor all cargo operations.

6. SUBJECT PROGRAM

	CONTENTS
1	1. Review of Maritime Transport. Seaborne Trade. World Fleet.
2	2. Bulk Cargoes. Bulk cargoes and bulk carriers. BLU IMSBC Coal, iron ore, bauxite, grain...
3	3. Containerized trade. Container seaborne trade Containers Container carriers. Container terminals Stowage and lashing Roro carriers. Car carriers.
4	4. Dangerous goods. IMDG. Classification Packing and tank provisions Consignment procedures Construction and testing of packagings, IBCs, tanks. Dangerous goods list. Special provisions and exceptions.
5	5. Liquid bulk cargoes. Crude oil tankers Product tankers Chemical tankers LPG tankers LNG tankers.

7. ASSESSMENT METHODS AND CRITERIA

Description	Type	Final Eval.	Reassessn	%
To write an essay about maritime transport: the transport of a determinate good on board a determinate kind of ship. 5 sheets maximum.	Work	No	No	5,00
Liquid cargo simulator work.	Laboratory evaluation	No	Yes	35,00
bulk cargoes, container cargoes and dangerous goods: test exam.	Activity evaluation with Virtual Media	No	No	25,00
bulk cargoes, container cargoes and dangerous goods: practice exercise.	Others	No	Yes	35,00
TOTAL				100,00

Observations

If you haven't passed prior exams you must do a final exam on simulator.

Observations for part-time students

Final exam on simulator and practice exercise.

8. BIBLIOGRAPHY AND TEACHING MATERIALS

BASIC

GENERAL.
Reglamentos.
OMI. SOLAS. Edición refundida de 2022. OMI. Londres, 2022.
OMI. MARPOL. Edición consolidada de 2022. OMI. Londres, 2022.
http://www.imo.org/InfoResource/mainframe.asp?topic_id=783
INTERNATIONAL CODE FOR THE SAFE CARRIAGE OF GRAIN IN BULK (International Grain Code) (1991 Edition) IMO
IMDG CODE. Ed. 2022. IMO.
BLU Code. Ed. 2011. IMO.
IMSBC Code. Ed. 2022. IMO.
CODE OF SAFE PRACTICE FOR CARGO STOWAGE AND SECURING (CSS Code)
(2022 Edition) IMO.
INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
DANGEROUS CHEMICALS IN BULK (IBC Code) (2021 Edition) IMO
INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK (IGC Code) (2016 Edition): IMO.
Libros
Thomas, O. Thomas' stowage. 7º ed. Brown, son & Ferguson. Glasgow, 2020.
Bases de datos.
Lloyd's casualty archive Lloyd's Marine Intelligence Unit, SeaSearcher. http://www.lloydsmiu.com/lmiu/index.htm
OMI. IMO Vega database. Det Norseke Veritas (DNV). V 19.0. 2014
Luzón. Estiba de Cargas Sólidas. Ed. Cartamar. La Coruña, 2016.